

# WEB BASED E-WASTE AND BATTERY LIFECYCLE MANAGEMENT SYSTEM

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**Abstract** - The rapid increase in electronic device usage within universities and institutions has led to a significant rise in electronic waste generation. Devices such as computers, printers, UPS systems, and networking equipment rely on batteries whose efficiency decreases with repeated use and environmental variations. In many institutions, maintenance records are still handled manually, making it difficult to track battery condition and schedule timely replacements. This paper presents a web-based E-Waste and Battery Lifecycle Management System that supports centralized device registration, maintenance record storage, battery health assessment, alert generation, and lifecycle analytics. The platform provides dashboard-based visualization of device performance and waste trends, helping administrators improve maintenance planning and promote sustainable disposal practices.

**KeyWords:**E-Waste Management, Battery Lifecycle Monitoring, Device Health Tracking, Preventive Maintenance, Web Dashboard, Sustainable Disposal

## I. INTRODUCTION

In recent years, the rapid growth of electronic devices in universities, research laboratories, offices, and industrial environments has significantly increased the generation of electronic waste. Equipment such as computers, printers, uninterruptible power supply (UPS) systems, routers, laboratory instruments, and communication devices are now essential for daily academic and administrative operations. As these devices are used continuously, their internal components—especially batteries—gradually lose efficiency due to repeated charging cycles, heat exposure, aging, and prolonged operational stress. Battery degradation is one of the major challenges in institutional electronic asset management. A weak or damaged battery can lead to unexpected

shutdowns, data loss, interruption of laboratory experiments, and reduced performance of critical systems. If the health of these batteries is not monitored properly, institutions may face sudden equipment failures, higher maintenance costs, and increased electronic waste accumulation. At present, many educational institutions still depend on manual maintenance logs, spreadsheets, or paper-based service registers to track device condition and replacement history. These conventional methods are often difficult to manage when devices are distributed across multiple departments and laboratories. Manual monitoring also makes it hard to identify early signs of battery failure, resulting in delayed maintenance actions and improper disposal decisions. In addition to operational challenges, improper handling of discarded batteries and outdated electronic equipment creates serious environmental concerns. Batteries contain hazardous materials that can affect soil, water, and human health if not disposed of through proper e-waste channels. Therefore, institutions require a smart and organized approach to monitor device health, extend battery lifespan, and support responsible disposal practices. To overcome these limitations, this paper proposes a Web-Based E-Waste and Battery Lifecycle Management System that provides centralized supervision of institutional electronic assets. The platform enables device registration, location mapping, maintenance record storage, battery health evaluation, automated alert generation, and dashboard-based analytics. By maintaining lifecycle records and generating timely replacement notifications, the system helps administrators improve preventive maintenance planning and reduce avoidable failures. The proposed system also contributes to sustainable technology management by supporting data-driven decisions for battery reuse, replacement scheduling, and final disposal. This not only improves operational efficiency but also strengthens institutional .

## II. LITERATURE REVIEW

The increasing use of digital methods in addressing the issue of e-waste generation and monitoring the performance of devices is evident in the recent literature. For instance, Shibby and Wijegoda [1] emphasized the application of web-based solutions for organizing and recycling e-waste in more effective ways. The study by Armstrong et al. [2] considered the process of tracking the transport of e-waste from its sources to places where recycling takes place. Strategies aimed at solving problems related to e-waste management were suggested by Suma [3] and Spoorthi [4]. They designed web-based systems for processing information about e-waste, which did not incorporate monitoring functions. In order to tackle emerging challenges, Kumar et al. [5] used IoT technologies for monitoring levels of e-waste, but they had to deploy additional hardware in order to achieve the goal. Device management and monitoring were examined separately in the studies by Ravi et al. [6] and Patel et al. [7], who focused on the maintenance of devices and processing e-waste data on cloud platforms. Singh et al. [8] stressed the importance of monitoring battery performance with rule-based techniques. Life cycle monitoring and sustainability were also studied in detail by Rahman et al. [9]. The authors suggested models to determine the need for replacement based on consumption behavior. Data analysis tools were employed by Chen et al. [10] to study e-waste creation trends and improve decision-making processes. Battery monitoring systems that employed Internet-of-Things (IoT) technologies, such as the system designed by Li et al. [11], ensured reliable real-time information but led to high costs. Literature review Predictive models have also been considered in the literature. Machine learning predictive models were used by Zhang et al. [12] for predicting device failure, whereas Ahmed et al. [13] focused on improving asset tracking via web-based approaches. Alert systems to notify users in case of failure, suggested by Verma et al. [14] and Khan et al. [22] have also been used recently. Multi-functional approaches with tracking, reporting and monitoring components were developed by Joseph et al. [15] and Iyer et al. [23]. Dashboard-driven systems for performance and ewaste trend visualizations were also created by Nair et al. [16] and Sharma et al. [24]. Furthermore, centralized data management and cloud computing were emphasized by Gupta et al. [17] and Mehta et al. [21]. Battery monitoring and failure prediction were also addressed using embedded system model and statistical approach respectively. The former was proposed by Hassan et al. [18] for real-time monitoring, whereas the latter was suggested by Roy et al. [19]. Low-cost models for smaller organizations were proposed by Das et al. [20]. From the literature review it is clear that most existing systems are focusing on one aspect of the problem

## III. SYSTEM ARCHITECTURE

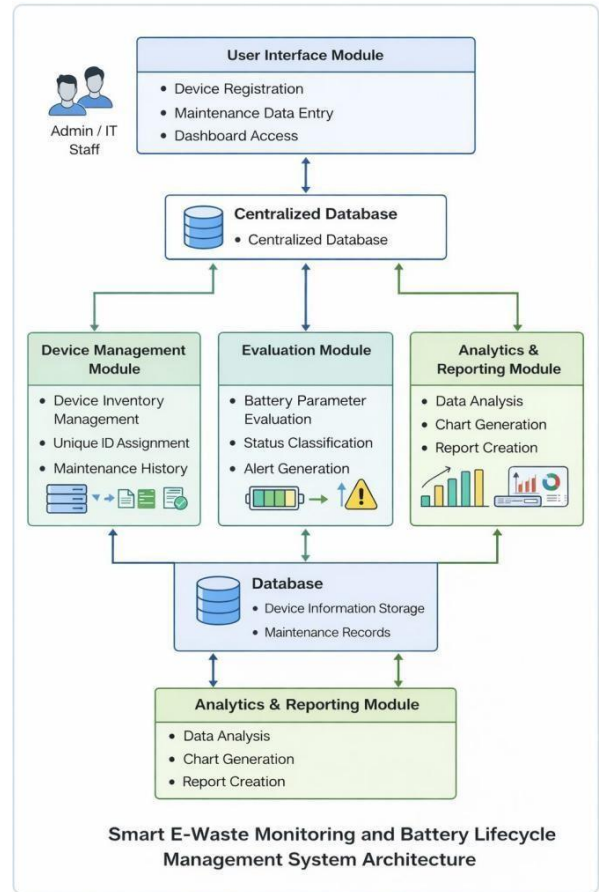


Fig 1: System Architecture

There are four major layers in this Architecture:

### 3.1 User Interface Layer

It provides user interface for administrators to enter details about devices and their maintenance.

### 3.2 Data Management Layer

It includes storage of all historical information about the devices in one database.

### 3.3 Evaluation Engine

It evaluates battery parameters according to the threshold logic.

### 3.4 Analytics Layer

It presents replacement rates of batteries, electronics waste accumulation rate, and failures rate. This layered architecture allows the application to be easily scalable and centrally controlled.

#### IV. METHODOLOGY

The suggested architecture is based on the methodology, which provides an effective way to monitor electronic devices and batteries throughout their lifecycle in a university environment. It comprises a number of stages such as device registration, maintenance data entering, battery evaluation, generating alerts, and analytics.

Admin Page:

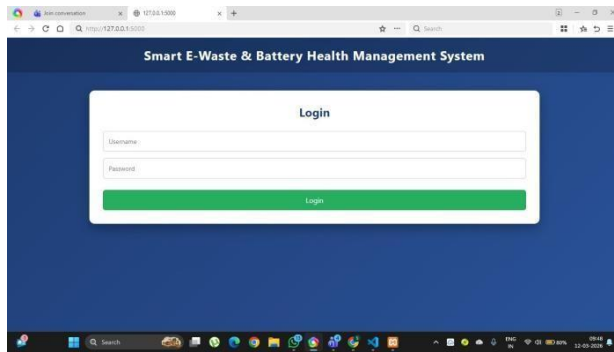


Fig 2:Admin Page

##### 4.1 Location Identification:

Figure 3 of the system entails the identification and registration of all operational sites in the institution. Some of these sites include computer laboratories, research laboratories, library systems, departmental sites, administrative offices, and IT equipment maintenance rooms. The administrator carries out the process at the time of installation of the system. By definition of these sites, it makes it easier to organize the management of devices within the various sites and check their operational capabilities.

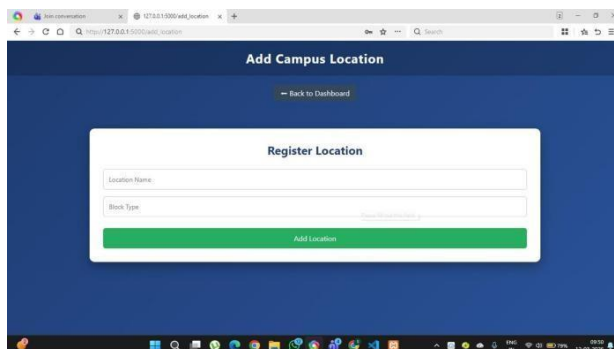


Fig 3: Location Page

##### 4.2 Device Registration:

Following the definition of the site where the equipment will be installed, fig 4 shows that all electronic devices used in the institution are registered. Information about each of the devices registered is recorded in terms of the type of device, installation

date, battery information, and location. Once the device is registered, a digital identifier number for the device is generated by the system.

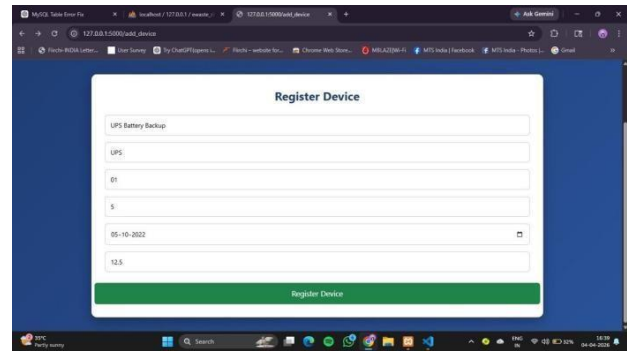


Fig 4:Device Registration Page

##### 4.3 Entering Data For Periodic Maintenance:

Periodically at weekly or monthly intervals, IT maintenance staff members enter certain device parameters in the system. Such parameters consist of battery voltage, operating temperature, and charge cycle status. It is noted that the system uses manual entries rather than hardware sensors, making it cheaper to implement. However, this system can still manage to monitor the state of operation of the devices through existing maintenance data records that figure in fig 5.

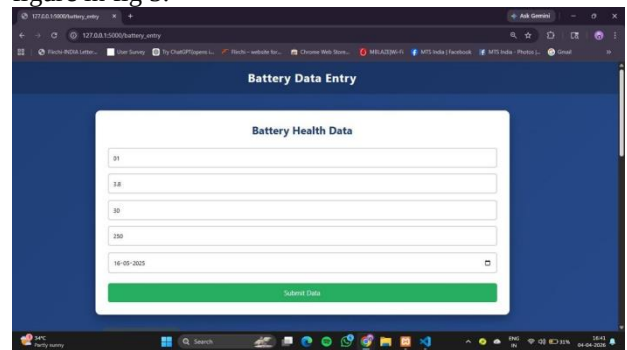


Fig 5:Data Entry Page

##### 4.4 Condition Analysis of Batteries:

Once data from the maintenance process have been entered, in fig 6, the system conducts condition analysis for the batteries on the basis of certain rule-based calculations. This analysis assesses the state of healthiness of the battery. On the basis of these calculated figures, the system categorizes the state of devices in different conditions such as normal condition, monitoring state, requiring replacement, or disposal stage.

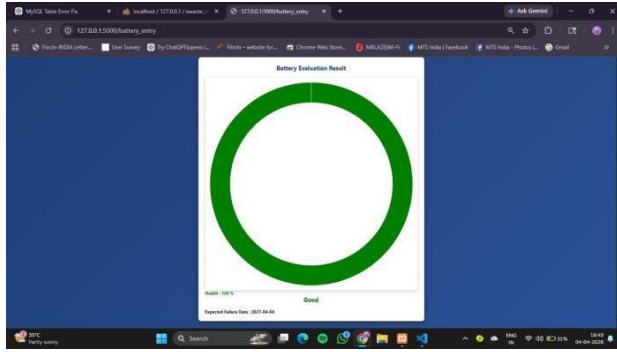


Fig 6: Battery Evaluation

#### 4.5 Device Information Storage & Lifecycle Tracking:

All data related to device information, maintenance, and evaluations is entered into a database where it is kept for future reference. The database stores the historical record of all registered devices so that their lifecycles can be tracked easily. The information stored also helps administrators recognize usage habits, maintenance, and replacement needs for the devices. fig7 & fig8 have been used for storing the collected data.

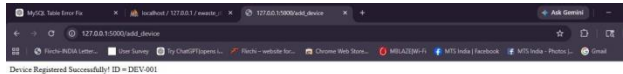


Fig 7:Device Data Storage

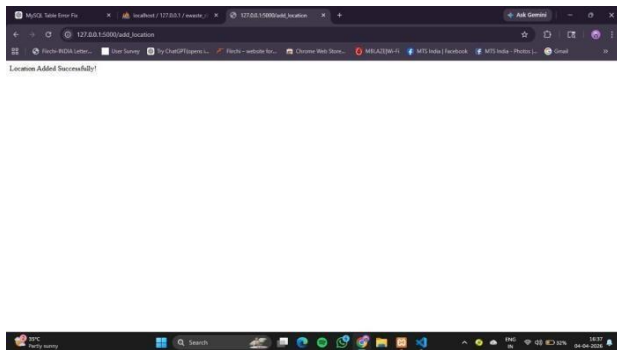


Fig 8:Location Data Storage

#### 4.6 Analytics and Report Generation:

The final step in fig 9 is the creation of analytical reports on data that has been stored. This is done by using charts and graphs to summarize data regarding the trends in battery replacements, performances of devices, and generation of e-waste.

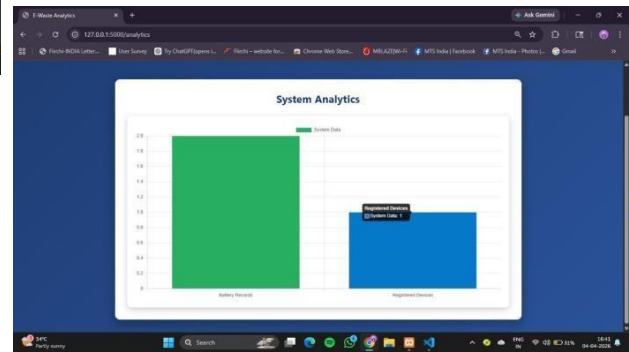


Fig 9:Analytics Generation

## V. RESULT AND ANALYSIS

The Fig 10, Fig 11 are the proposed Smart E-Waste Monitoring and Battery Lifecycle Management System, which is designed for the purpose of monitoring electronic devices and analyzing their battery condition within an institution. This system was tested by recording numerous electronic devices, including computers, UPS and printers across different departments like laboratories and administrative offices. Maintenance-related information like voltage, temperature, and charge cycle data was periodically updated in the system. Using this data, the evaluation tool analyzed the battery status using rule-based calculation. The system has been able to successfully classify electronic devices based on their performance in different stages including normal operation, under monitoring stage, and require replacement. The analytics tool generated graphs that were used to report battery status trend, failure rate per month and e-waste generation trend. Such graphs assisted the administrators in properly identifying the condition of the devices, those requiring maintenance, and replacement. This indicates that the proposed system has proven its usefulness in monitoring and evaluating electronic device performance, hence making it easy for administrators to perform proper maintenance and manage e-waste generation in institutions. This system also makes it easy for the institution to promote sustainability by effectively managing

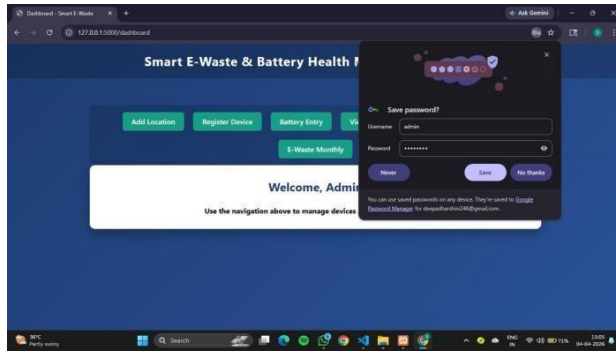


Fig 10:User Page

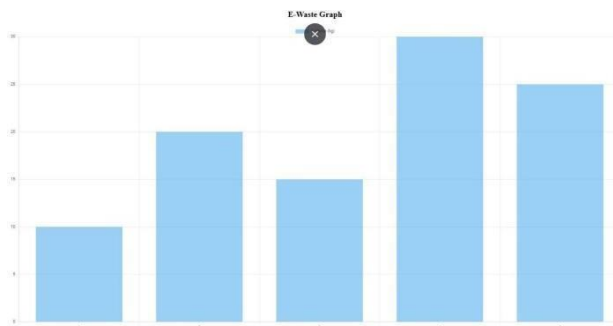


Fig 11:E-Waste Graph Genrate

## VI. FUTURE WORK

Moreover, the above proposed system can be enhanced by incorporating the use of IoT sensors to automatically record data on the performance of the battery such as the battery's voltage, temperature, and charge cycle counts. This will increase data accuracy. In the future, the application of machine learning algorithms can enable the prediction of future battery failures using past maintenance records. The platform can also be enhanced with the capability of classifying ewaste into reusable, recyclable, and disposable ewaste based on smart classification. The use of cloud computing can make the supervision more scalable since cloud computing enables the tracking of batteries across multiple institutions in terms of campuses and branch offices. Cloud storage will also facilitate improved backups of the recorded data. Lastly, the above system can be enhanced with the incorporation of smart ewaste classification modules, which will help in sorting ewaste into either reusable, recycling or disposal categories. All in all, the future developments can make the proposed rule-based system into a fully intelligent and automated life cycle supervision platform.

## VII. CONCLUSION

This paper has demonstrated a Web-Based E-Waste and Battery Lifecycle Management System capable of enhancing institutional management and monitoring of devices and batteries. Indeed, this system has been able to provide the integration of registration, mapping, battery parameters evaluation, alert, recording, and analysis of the device lifecycle information. The system overcomes the weaknesses associated with manual maintenance records by enhancing visibility regarding battery status in laboratories, offices, and academic departments within an institution. The use of rules-based battery evaluation will enable early detection of battery problems, and thereby trigger timely alerts for replacement, thus minimizing unplanned breakdowns. The analytics dashboard will assist in decision making with regard to wear trend of the battery, frequency of battery replacements, and departmental e-waste statistics. In summary, the management system is an efficient and effective solution for improving operations, maintenance planning, asset management, and sustainable e-waste management.

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